

Empirical study on usability and user experience of vehicle-mounted AR-HUD

Jin-Long Lin

Doctoral Program in Design,
College of Design, National Taipei Uni-
versity of Technology
Taipei, Taiwan
t110859406@ntut.org.tw

Meng-Cong Zheng

Department of Industrial Design,
National Taipei University of
Technology
Taipei, Taiwan
zmcdesign@mail.ntut.edu.tw

Chen-Rao Zhong

Doctoral Program in Design,
College of Design, National Taipei Uni-
versity of Technology
Taipei, Taiwan
t110859407@ntut.org.tw

Abstract—This study mainly discusses AR-HUD availability for three mainstream models in the Chinese market (Benz EQS, VW ID.6 X, and EXEED RX). We invited eight people working in automotive-related fields to conduct heuristic evaluations, questionnaires, and in-depth interviews. After evaluation, we found that the VW ID.6 X and EXEED RX models had many usability problems in the three dimensions of function, vision, and emotion. In addition, according to the questionnaire results, Benz EQS performed best in terms of willingness to use, sense of experience, sense of security, and aesthetics. There is no significant difference in the subjective evaluation of the AR-HUD of the VW ID.6 X and EXEED RX models. Benz EQS is favored by everyone for its excellent display technology and visual design effects.

Keywords—AR-HUD, usability, user experience

I. INTRODUCTION

Augmented reality (AR) is a graphics technology that combines computer-generated visual information with real-world scenes. Augmented reality-head-up display (AR-HUD) is an emerging technology that applies augmented reality technology to automobile head-up display systems. As early as 1999, General Motors began to develop vehicle head-up displays (HUD), and AR technology was first used in vehicles in 2010 [1].

AR-HUD may enhance the driver's attention to the road and reduce the cognitive workload while driving [2]. However, as a new driving experience, AR-HUD may also distract the driver's attention and present an interface that is difficult to read, thereby increasing unsafe factors [3]. Therefore, this study focuses on analyzing the current usage of mainstream models equipped with AR-HUD. Through usability testing, design suggestions for improving AR-HUD were drawn to improve user acceptance and driving safety.

II. RESEARCH METHOD

Regarding research methods, we used the heuristic evaluation and questionnaire method. Among them, in the heuristic evaluation, we provided participants (experts) with an AR-HUD design guide (see TABLE I). The guide is built with reference to previous usability testing work [4] and in conjunction with in-vehicle evaluation. The guide includes three dimensions: functional, visual, and emotional, with a total of 12 design standards. After the heuristic evaluation, we invited experts to conduct a questionnaire survey, which took the form of a 5-point Likert scale. Some of the questions in the questionnaire refer to the System Usability Scale (SUS) [5]. The questionnaire includes scores on six facets: Willingness to use, Sense of experience, Ease of learning, de-redundancy, Sense of security, and Aesthetics. Also, we let experts know that the study solely focuses on AR-HUD evaluation, excluding car brand and performance.

TABLE I. AR-HUD GUIDELINES IN HEURISTIC EVALUATION

Dimensions	Facet	Content (brief description)
Function	Completeness	Whether the AR-HUD contains necessary display information.
	Effectiveness	Whether AR-HUD displays current information accurately.
	Efficiency	Whether AR-HUD displays current information efficiently.
	Stability	Whether the AR-HUD runs smoothly and without crashes.
Vision	Brightness	Whether the AR-HUD is glaring or has low brightness.
	Clarity	Whether the various information of AR-HUD is clearly displayed.
	Color temperature	Whether the AR-HUD screen is white or yellowish, etc.
	Readability	Whether the icon/text design of AR-HUD is easy to understand.
Emotion	Dizziness	Does dizziness occur due to AR-HUD shake/ghosting?
	Interesting	Is the AR-HUD design vivid and exciting?
	Safety	Do you feel safer using AR-HUD?
	Satisfaction	Overall satisfaction with AR-HUD.

III. EXPERIMENTAL EQUIPMENT AND PROCEDURES

A. AR-HUD Equipment

This experiment selected three models of Benz EQS, Volkswagen ID.6 X, and Chery EXEED RX equipped with AR-HUD as the evaluation objects. TABLE II and Fig 1 show the AR-HUD parameters and illustrations of these three cars.

TABLE II. AR-HUD PARAMETERS OF THE CAR IN THE EXPERIMENT

Auto	FOV (°)	VID (m)	Projection (inches)	Imaging method
Benz EQS	10*5	10	77	DLP
VW ID.6 X	9*4	10	70.9	TFT
EXEED RX	10*3	7.5	50	TFT



Fig. 1. AR-HUD interface for three cars

B. Experimental Design

We have pre-designated a fixed driving route, which includes highways and urban roads, with a total route of 20 kilometers. The evaluation time for each car is approximately 30 minutes. Each expert took turns driving and using three vehicles equipped with AR-HUD and conducted heuristic evaluations and questionnaire surveys.

C. Participants

The eight experts who participated in this heuristic evaluation were all technical engineers (4) and interface designers (4) responsible for AR-HUD development. Their average working age is 5.88 (SD=1.553).

IV. TEST RESULTS AND ANALYSIS

After usability testing, we obtained interview text and questionnaire score data. Subsequently, we used IBM SPSS 26.0 statistical software to conduct a one-way repeated measures ANOVA analysis of the questionnaire score.

A. Heuristic evaluation results

In the heuristic evaluation, each expert evaluated the AR-HUD of each car in three dimensions. In the dimension of the function, compared with VW ID.6 and EXEED RX, the Benz EQS lacks snow functions. In addition, VW ID.6 and EXEED RX are more flexible in HUD settings. The HUDs of these two cars can independently cancel information such as navigation, assisted driving, and traffic signs. However, Benz EQS has 5 HUD modes: minimalist, sports, standard, AR, and energy-saving. In addition, the Benz EQS automatically detects the driver's head position and instantly adjusts the AR display position on the HUD.

In the visual dimension, experts agree that the Benz EQS's AR-HUD is clearer and more transparent than the other two cars. The clarity of the VW ID.6's AR navigation display is the worst, and the AR navigation arrows cannot even be seen under strong light and special backgrounds. In addition, some icons and text in the AR-HUD interfaces of Benz EQS and VW ID.6 are too small and difficult to read. The AR navigation instructions of EXEED RX could be better. Because its AR indication is different from that of an actual road situation, and the AR arrow is too small. In the emotional dimension, the Benz EQS captivated participants with its outstanding dynamic effect design. Therefore, Benz EQS is also the best in terms of satisfaction.

B. Questionnaire Results

According to the one-way repeated measures ANOVA on the six facets of the questionnaire in TABLE III, willingness to use ($F_{2,14}=13.267$, $P=0.000 < 0.05$), sense of experience ($F_{2,14}=12.629$, $P=0.000 < 0.05$), sense of security ($F_{2,14}=6.635$, $P=0.009 < 0.05$) and aesthetics ($F_{2,14}=25.282$, $P=0.000 < 0.05$) have significant main effects.

TABLE III. ONE-WAY REPEATED MEASURES ANOVA FOR RESULTS

Facet	SS	df	MS	F	P	Post Hoc
Willingness to use	16.583	2	8.292	13.267	<0.001**	EQS > ID.6X; EQS > RX
Sense of experience	14.583	2	7.292	12.629	<0.001**	EQS > ID.6X; EQS > RX
Ease of learning	1.000	2	0.500	0.528	0.595	
de-redundancy	0.583	2	0.292	0.467	0.637	
Sense of security	9.083	2	4.542	6.635	0.009*	EQS > RX
Aesthetics	25.583	2	12.792	25.282	<0.001**	EQS > ID.6X; EQS > RX

* Significantly different at $\alpha = 0.05$ level (* $P < 0.05$), ** Sig at $\alpha = 0.01$ level (** $P < 0.01$).

Regarding the willingness to use AR-HUD, the Post Hoc test (LSD) results show that there is a significant difference between Benz EQS ($M=4.75$, $SD=0.463$) and VW ID.6X ($M=2.88$, $SD=0.991$) ($P=0.002 < 0.05$). There is also a significant difference between Benz EQS and EXEED RX ($M=3.13$, $SD=1.246$) ($P=0.006 < 0.05$). In the facet of AR-HUD experience, there is a significant difference between Benz EQS ($M=4.75$, $SD=0.463$) and VW ID.6X ($M=3.50$, $SD=0.756$) ($P=0.011 < 0.05$). There is also a significant difference between Benz EQS and EXEED RX ($M=2.88$, $SD=0.835$) ($P=0.000 < 0.05$). Regarding the sense of security of AR-HUD, there is a significant difference between Benz EQS ($M=4.13$, $SD=0.835$) and EXEED RX ($M=2.63$, $SD=0.916$) ($P=0.003 < 0.05$). In the facet of the aesthetics of AR-HUD, there is a significant difference between Benz EQS ($M=4.63$, $SD=0.518$) and VW ID.6X ($M=2.38$, $SD=0.744$) ($P=0.000 < 0.05$). There is also a significant difference between Benz EQS and EXEED RX ($M=2.50$, $SD=0.926$) ($P=0.000 < 0.05$). In other words, Benz EQS's AR-HUD is the best choice in terms of willingness to use, sense of experience, sense of security, and aesthetics.

V. CONCLUSION

From interviews with experts, we learned that the current AR-HUD has too much information in the interface, which may interfere with the driver's attention to the surrounding environment [6]. While the performance of the Benz EQS in our evaluation is commendable, the usability issues we listed in our heuristic evaluation also need to be considered. The reason why Benz EQS's AR-HUD performs better may be due to the application of Digital Light Processing (DLP) imaging technology on the one hand. On the other hand, more cool and interesting dynamic effects were added to its interface design (see Fig. 2). Benz EQS displays impactful indication effects on both the HUD and the central control screen during AR navigation. The AR-HUD of the other two models is stiffer and more mediocre in visual effects.



Fig. 2. AR-HUD dynamic effect display of Benz EQS

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